

PAPER 1
SECTION B: Question-Answer Book B

Time Allowed: 8:30 am – 10:30 am
(2 hours for both Section A and B)
This paper must be answered in English.

INSTRUCTIONS

- (1) After the announcement of the start of the examination, you should first write your Candidate Number in the space provided on Page 1 and stick the label in the space provided on Page 1.
- (2) This section consists of TWO parts, Parts I and II.
- (3) **ANSWER ALL QUESTIONS in both Parts I and II.** Write your answers in the spaces provided in this Question-Answer Book. Do not write in the margins. Answers written in the margins will not be marked.
- (4) Supplementary answer sheets will be provided on request. Write your candidate number and question number on each sheet.
- (5) No extra time will be given to candidates for sticking on the barcode labels or filling candidate number and question number after the 'Time is up' announcement.
- (6) The last page of the question-answer book contains a list of SQL commands and spreadsheet functions which you may find useful.

Please stick the barcode label here.

Candidate
Number

Marker's Use Only

Question
No.

Marks

1

2

3

4

5

6

7

8

9

Total

PART I

Answer all questions.

1. A secondary school annually assesses students' academic performance and discipline to determine whether to award scholarships. The table below shows some simplified samples of student data:

	A	B	C	D	E	F	G
1	Student number	English name	Academic performance (0-100 marks)	Discipline grade (A-D)	Participation in extracurricular activities (Y/N)	Eligibility in scholarship (Y/N)	Remarks
2	S001	Alice Chan	85	A	Y		
3	S002	Bob Wong	70	B	N		
4	S003	Carol Lee	92	A	Y		
5	S004	David Ng	65	C	Y		
6	S005	Emily Lam	78	C	N		
7	S006	Frank Ho	55	D	Y		
8	S007	Grace Ma	68	B	Y		
...	...	...	...	...	...	...	...
100	S099	Ivy Yeung		A	Y		

The scholarship evaluation criteria are as follows:

Basic requirements: Academic performance must be 75 marks or above.

Additional requirements: Conduct grade must be "A" or participation in extracurricular activity must be "Y".

If a student meets all the requirements, the eligibility in scholarship (column F) will display "Y"; otherwise, it will display "N".

- (a) Suggest a value in C100 so that "Y" is displayed in F100. _____ (1 marks)

- (b) A formula is used to determine the eligibility of scholarship in F2. Complete the formula for F2 below. (3 marks)

=IF (AND (_____ , OR (_____ , _____)) , "Y" , "N")

2. Mr. Chan is a digital content creator who frequently uses a laptop to edit video clips.

- (a) How does the graphics processing unit (GPU) in the laptop help Mr. Chan's work? (1 mark)

- (b) Mr. Chan is considering two different external secondary storage devices to store his artwork. List one disadvantage of each device. (2 marks)

	HDD	SSD
Disadvantage		

- (c) Mr. Chan has created a 1GB video. Should he use a cloud-based storage or an external secondary storage device? Explain briefly. (1 mark)

3. Currently, the government provides various services to citizens through mobile applications. However, this method may pose challenges for the elderly. A community center is assisting the elderly in learning to use smartphones to access government services.

- (a) Give two examples of mobile application design that are designed to optimize the user experience for the elderly. (2 marks)

- (b) Scammers often impersonate government officials to carry out fraud targeting elderly people.

- (i) Give an example of this type of scam. (1 mark)

- (ii) Suggest a way to reduce the number of elderly people falling into such scams. (1 mark)

4. Peter is a programmer. He has created a website, www.peter.com.hk, to share his open-source program with registered users.

- (a) A user finds that both the URLs <http://www.peter.com.hk> and <https://www.peter.com.hk> can access the same webpage. What is the difference between the two URLs? Briefly explain. (2 marks)

- (b) A user submits a password to log in Peter's website. Explain how Public Key Infrastructure (PKI) ensures the confidentiality of the password during transmission. (2 marks)

- (c) John modifies the source code of a program downloaded from Peter's website and then sells it as commercial software. Does John violate the copyright of Peter? Briefly explain your answer. (1 mark)

5. A large shopping mall has implemented a smart parking system, allowing drivers to easily monitor the availability of parking spaces. Sensors have been installed in each parking space to identify whether it is occupied or not, transmitting the signals back to a central computer system. In addition, the mall has a central server to process all vehicle entry and exit records and provide online reservation services.

- (a) What is the minimum number of bits required to represent the data returned by the sensor? Explain briefly. (2 marks)

- (b) Suggest a suitable type of processing system for handling online reservations. Explain briefly. (1 marks)

- (c) The administrator noticed a significant slowdown in the response of the central server when many vehicles enter and exit the carpark simultaneously during peak hours at weekends. Suggest a utility program to check this issue. (1 mark)

6. Martin is designing a website for a school. He has two image files.

- **Image A:** A high-resolution color photograph of the campus.
- **Image B:** The school logo, which is composed of geometric shapes in a circular form.

Suggest the most suitable file format (JPG, PNG, BMP or SVG) for each image and provide explanation. (4 marks)

	File format	Explanation
Image A		
Image B		

PART II

Answer all questions.

7. A school in a certain country is implementing a "Smart Library System" to provide students with access to digital resources via the school network. The system includes new computers, a wireless and wired network, and Internet-based services. The school is concerned about hardware performance and network security.

- (a) Give **one** example of digitization that occurs in a library. (1 mark)

- (b) Students can use a wireless network to access digital resources in the Smart Library System.

- (i) When streaming digital resources to wireless devices, TCP/IP is involved. Briefly describe how data is transmitted with reference to the concepts of data packets. (3 marks)

- (ii) Suggest **two** security measures the school should implement to protect the Smart Library System from unauthorized access. (2 marks)

(c) The Smart Library System uses access points to provide network access for student devices.

(i) State the main function of access points. (1 mark)

(ii) Suggest **two** factors the school should consider when deploying access points to ensure reliable coverage throughout the library. (2 marks)

(d) The school is considering using AI-powered technology to allow students to borrow and return books without physical library cards.

(i) Suggest **one** biometric authentication technology that could be used for this purpose. (1 mark)

(ii) Describe a potential threat related to implementing the technology in the library. (1 mark)

8. A database is used in a vending machine system to store information on the products sold. The data is recorded in the data table `ITEM`.

`ITEM`

Field name	Data type	Description	Example
ID	character	Product Identification Code	C001
NAME	character	Product name	Good Taste Candy
PRICE	number	Selling price	12.50
CQ	integer	Current quantity in the slot	32
MC	integer	Maximum capacity of the product	50
VM	X	Serial number of vending machine	001

- (a) Describe **one** validation check performed on `CQ` apart from type check and existence check. (1 mark)

- (b) Suggest the data type for `VM`. Explain briefly. (2 mark)

- (c) Suppose that there are seven records in `ITEM` as follows:

ID	NAME	PRICE	CQ	MC	VM
F001	Hibi senbei	8.00	25	80	001
F002	Spicy potato chips	18.50	18	30	002
F003	Little bear biscuit – jumbo size	18.00	20	39	002
F004	Milk chocolate bar	10.00	35	50	002
F005	Cheese-flavored sandwich biscuit	8.50	22	35	001
F006	Peanut butter biscuit	9.50	15	25	001
F007	Chocolate biscuit	6.50	30	40	001

What is the output after executing the following SQL statement?

(i) `SELECT VM, NAME, PRICE FROM ITEM
WHERE PRICE <= 10
ORDER BY VM, PRICE`

(1 mark)

(ii) `SELECT VM, AVG(CQ) FROM ITEM
WHERE PRICE > 8 AND NAME LIKE '%biscuit%'
GROUP BY VM;`

(1 mark)

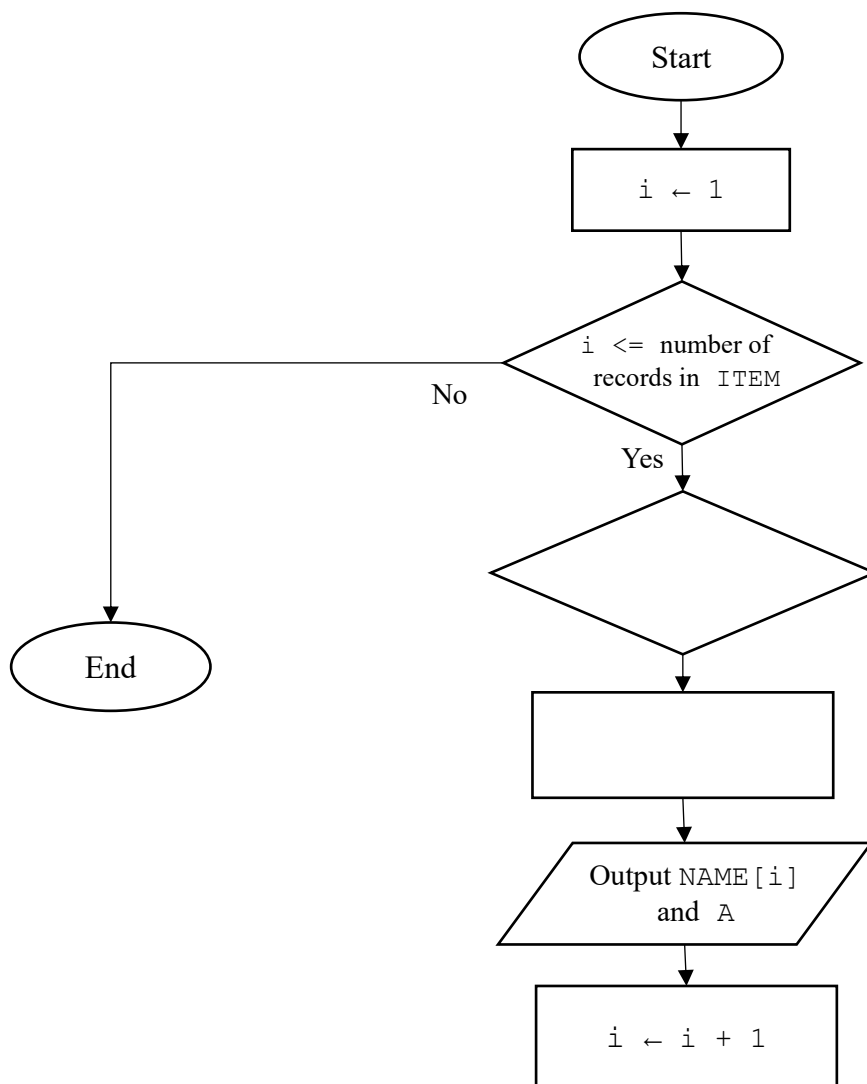
The programmer needs to design a module to determine whether a specific item in a vending machine needs to be restocked. The data is exported from database for further processing.

It is known that the data in the fields `NAME`, `CQ` and `MC` in the table `ITEM` has been stored in parallel arrays `NAME`, `CQ` and `MC` in the module respectively. Using the data in part (c) as an example, the fields in the first record are stored as follows:

```
NAME[1] = "Hibi senbei"  
CQ[1] = 25  
MC[1] = 80
```

The system needs to check the current stock of all products in the vending machine. If the current stock is less than or equal to 20% of the maximum capacity, the system will prompt that the product needs to be refilled, and the amount `A` that needs to be refilled to the maximum capacity.

- (d) In the flowchart below, the variable `i` is used to store the array index. Based on the above refill requirements, fill the content in the blank shapes. Add appropriate connecting lines and arrows to display the correct logic flowchart. (4 marks)



- (e) To ensure the modules work correctly, the following tests will be conducted. Assume the maximum capacity is **50**. Fill in the table below. (3 marks)

Test	Recommended value of an element in CQ
Test normal situation of refill	
Test the boundary value	
Test invalid data	

Please tick the appropriate box to indicate the programming language used in Question 9.

Python ☐

C++ ☐

Pascal ☐

9. A programmer is writing a monitoring system for an "all-weather smart greenhouse". This system records the temperature (in degrees Celsius °C) in different days of the week and stores it in a list called `temperatures` .

The system has collected the following data from sensors throughout the week:

```
temperatures = [24.5, 26.0, 28.5, 31.2, 29.0, 33.5, 27.0]
```

- (a) Explain why the system uses an array instead of a variable to store temperature records.(2 marks)

- (b) The programmer wrote the following code to calculate the weekly average temperature of the greenhouse but found that the result was incorrect.

PYTHON
<pre>total = 0 count = 0 for i in range(0, 7): total = total + temperatures[i] count = count + 1 average = total / count + 1</pre>

C++
<pre>double total = 0; int count = 0; for (int i=0; i<7; i++) { total = total + temperatures[i]; count = count + 1; } double average = total / count + 1;</pre>

Pascal
<pre> var total, average, t: Real; count: Integer; begin total := 0; count := 0; for i := 1 to 7 do begin total := total + temperatures[i]; count := count + 1; end; average := total / count + 1; end; </pre>

- (i) Identify the type of error that occurred in the above code. (1 mark)
-
- (ii) Correct the code in that line to calculate the average correctly. (1 mark)
-
- (c) The greenhouse manager needs to determine the highest temperature of the week. Please write a program using Python / C++ / Pascal in the blank space below to accomplish this task and store the highest temperature in the variable `max_temp`. (Note: Built-in functions cannot be used to implement the algorithm.) (4 marks)

- (d) To protect the plants, the system needs to issue a warning if the temperature exceeds 30°C. Please write a program using Python / C++ / Pascal to calculate the total number of readings exceeding 30°C and store the total number in the variable `high_temp_count` . (3 marks)

The programmer decided to write the code in different modules.

- (e) Describe **one** advantage of using modularization. (1 mark)

END OF PAPER

Database (SQL commands)

Constants	TRUE, FALSE
Operators	+, -, *, /, >, <, =, >=, <=, <>, %, _ , ' , AND, NOT, OR
SQL	AVG, MAX, MIN, SUM, AS, BETWEEN, BY, ASC, DESC, DISTINCT, FROM, GROUP, HAVING, LIKE, NULL, ORDER, SELECT, WHERE

Spreadsheet

Constants	TRUE, FALSE
Operators	+, -, *, /, <, >, =, <>, <=, >=, &
Functions	INT, RAND, SQRT, ROUND, AND, NOT, OR, LEFT, LEN, MID, RIGHT, AVERAGE, COUNT, COUNTIF, MAX, MIN, RANK, SUM, SUMIF, FIND, XLOOKUP, IF